

DCL (Data Control Language)

DCL ka use **users ko permissions (privileges) dene aur wapas lene** ke liye hota hai.

DCL ke main commands:

1. GRANT
 2. REVOKE
-

1. GRANT

Definition

Kisi user ko database ya table par **permission dena**.

</> SQL

```
GRANT permission  
ON database.table  
TO 'username'@'localhost';
```

A. SELECT Permission

User sirf data dekh sakta hai.

</> SQL

```
GRANT SELECT  
ON college.student  
TO 'tarun'@'localhost';
```

B. INSERT Permission

SQL

```
GRANT INSERT
ON college.student
TO 'tarun'@'localhost';
```

C. UPDATE Permission

SQL

```
GRANT UPDATE
ON college.student
TO 'tarun'@'localhost';
```

D. DELETE Permission

SQL

```
GRANT DELETE
ON college.student
TO 'tarun'@'localhost';
```

E. Multiple Permissions

SQL

```
GRANT SELECT, INSERT, UPDATE
ON college.student
TO 'tarun'@'localhost';
```

F. All Permissions

SQL

```
GRANT ALL PRIVILEGES  
ON college.student  
TO 'tarun'@'localhost';
```

G. Entire Database

SQL

```
GRANT ALL PRIVILEGES  
ON college.*  
TO 'tarun'@'localhost';
```

H. Entire Server

`</> SQL`

```
GRANT ALL PRIVILEGES  
ON *.*  
TO 'tarun'@'localhost';
```

Means

Sabhi databases

WITH GRANT OPTION

User dusre users ko bhi permission de sakta hai.

SQL

```
GRANT SELECT
ON college.student
TO 'tarun'@'localhost'
WITH GRANT OPTION;
```

2. REVOKE

Definition

Di hui permission wapas lena.

Syntax

SQL

```
REVOKE permission
ON database.table
FROM 'username'@'localhost';
```

A. Remove SELECT

SQL

```
REVOKE SELECT  
ON college.student  
FROM 'tarun'@'localhost';
```

B. Remove INSERT

SQL

```
REVOKE INSERT  
ON college.student  
FROM 'tarun'@'localhost';
```


C. Remove UPDATE

⟨⟩ SQL

```
REVOKE UPDATE  
ON college.student  
FROM 'tarun'@'localhost';
```

D. Remove DELETE

⟨⟩ SQL

```
REVOKE DELETE  
ON college.student  
FROM 'tarun'@'localhost';
```

E. Remove Multiple Permissions

SQL

```
REVOKE SELECT, INSERT
ON college.student
FROM 'tarun'@'localhost';
```

F. Remove All Permissions

SQL

```
REVOKE ALL PRIVILEGES
ON college.student
FROM 'tarun'@'localhost';
```

MySQL 8.0 me pehle user create karna padta hai, phir us user ko permissions di jaati hain.

SQL

```
CREATE USER 'tarun'@'localhost'
IDENTIFIED BY 'Tarun@123';

GRANT SELECT, INSERT
ON college.student
TO 'tarun'@'localhost';
```

TCL (Transaction Control Language)

TCL ka use transaction ko **start, save, commit** ya **rollback** karne ke liye hota hai.

TCL Commands

1. `START TRANSACTION` (ya `BEGIN`)
2. `COMMIT`
3. `ROLLBACK`
4. `SAVEPOINT`
5. `ROLLBACK TO SAVEPOINT`
6. `RELEASE SAVEPOINT`
7. `SET AUTOCOMMIT`

Transaction Kya Hai?

Transaction = SQL statements ka ek group jo **ek unit** ki tarah execute hota hai.

Example:

ATM Transfer

Account A (-1000)

↓

Account B (+1000)

Dono statements successful honi chahiye.

Agar ek fail ho jaye to dono cancel.

Example

SQL

```
START TRANSACTION;
```

```
UPDATE student  
SET marks=95  
WHERE id=1;
```

```
UPDATE student  
SET marks=90  
WHERE id=2;
```

3. ROLLBACK

Saare changes undo.

SQL

```
ROLLBACK;
```

Example

SQL

```
START TRANSACTION;
```

```
UPDATE student  
SET marks=100  
WHERE id=1;
```

```
ROLLBACK;
```

SQL

```
START TRANSACTION;
```

```
UPDATE student  
SET marks=90  
WHERE id=1;
```

```
SAVEPOINT sp1;
```

```
UPDATE student  
SET marks=80  
WHERE id=2;
```

```
SAVEPOINT sp2;
```

```
UPDATE student  
SET marks=70  
WHERE id=3;
```

</> SQL

```
START TRANSACTION;
```

```
UPDATE student  
SET marks=90  
WHERE id=1;
```

```
SAVEPOINT sp1;
```

```
UPDATE student  
SET marks=80  
WHERE id=2;
```

```
UPDATE student  
SET marks=70  
WHERE id=3;
```

```
ROLLBACK TO sp1;
```

```
COMMIT;
```

7. AUTOCOMMIT

Check

SQL

```
SELECT @@autocommit;
```

Disable

SQL

```
SET autocommit=0;
```

Enable

SQL

```
SET autocommit=1;
```


SQL

```
SET autocommit=0;
```

```
START TRANSACTION;
```

```
UPDATE student  
SET marks=90  
WHERE id=1;
```

```
SAVEPOINT A;
```

```
UPDATE student  
SET marks=80  
WHERE id=2;
```

```
SAVEPOINT B;
```

```
UPDATE student  
SET marks=70  
WHERE id=3;
```

```
ROLLBACK TO B;
```

Bank Transfer

SQL

```
START TRANSACTION;
```

```
UPDATE account  
SET balance=balance-1000  
WHERE acc_no=101;
```

```
UPDATE account  
SET balance=balance+1000  
WHERE acc_no=102;
```

```
COMMIT;
```

Failure

SQL

```
START TRANSACTION;
```

```
UPDATE account  
SET balance=balance-1000  
WHERE acc_no=101;
```

-- Error

```
ROLLBACK;
```

Command	Purpose
START TRANSACTION	Transaction start karta hai
COMMIT	Changes permanently save karta hai
ROLLBACK	Changes undo karta hai
SAVEPOINT	Checkpoint banata hai
ROLLBACK TO SAVEPOINT	Savepoint tak rollback karta hai
RELEASE SAVEPOINT	Savepoint hata deta hai
AUTOCOMMIT	Automatic commit on/off